TMUA / WORKED SOLUTIONS

Set C Paper 1

Applications of Mathematical Knowledge 20 questions

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Inequalities

1. For which real values of x is the following inequality true?

$$x^2 < (x-3)^2 < (2x+1)^2$$

A $x < -4$ or $\frac{2}{3} < x < \frac{3}{2}$

B $x < -4$ or $x > \frac{2}{3}$

C $-4 < x < \frac{2}{3}$

D $-4 < x < \frac{3}{2}$

E $\frac{2}{3} < x < \frac{3}{2}$ or $x > 4$

Solution **Answer A**

The chain is true only when both inequalities hold:

$$x^2 < (x-3)^2$$

$$(x-3)^2 < (2x+1)^2$$

First solve the left-hand inequality. Expanding and simplifying,

$$x^2 < x^2 - 6x + 9 \implies 6x < 9 \implies x < \frac{3}{2}$$

Now solve the second inequality. Rearranging,

$$(2x+1)^2 - (x-3)^2 > 0$$

Using the difference-of-two-squares identity $a^2 - b^2 = (a-b)(a+b)$,

$$\big[(2x+1) - (x-3)\big]\big[(2x+1) + (x-3)\big] > 0$$

$$(x+4)(3x-2) > 0$$

Using difference-of-two-squares here is optional: expanding the squared expressions and then factorising gives the same result.

A product is positive when its factors are both positive or both negative. For both factors to be positive,

$$x > -4 \quad \text{and} \quad x > \frac{2}{3} \implies x > \frac{2}{3}$$

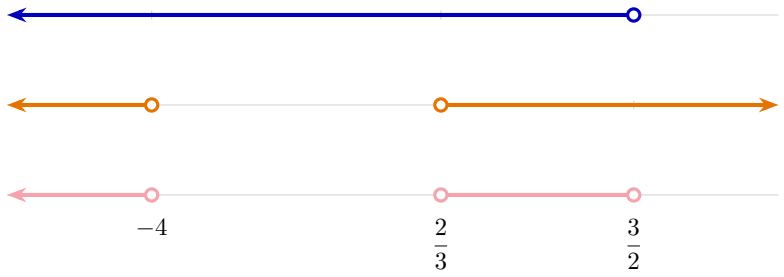
For both factors to be negative,

$$x < -4 \quad \text{and} \quad x < \frac{2}{3} \implies x < -4$$

Thus the second inequality holds when

$$x < -4 \quad \text{or} \quad x > \frac{2}{3}$$

Finally, take the overlap with $x < \frac{3}{2}$. On the aligned number lines below, blue shows the first inequality, orange the second, and pink their overlap. The marked values are not drawn to scale.



The overlap is

$$\boxed{x < -4 \quad \text{or} \quad \frac{2}{3} < x < \frac{3}{2}}$$

All endpoints remain excluded because the inequalities are strict. The correct option is **A**.

Alternative Solution:

Test values that distinguish the proposed sets: 0 lies in the central ranges, 2 tests the upper ranges, and -5 lies below -4.

For $x = 0$, the original chain evaluates to

$$0 < 9 < 1$$

This is false. Options C and D both include 0, so eliminate them.

For $x = 2$, the original chain evaluates to

$$4 < 1 < 25$$

This is false. Option B includes 2 because $2 > \frac{2}{3}$, so eliminate B.

For $x = -5$, the original chain evaluates to

$$25 < 64 < 81$$

This is true. Of the remaining options, A includes -5, whereas E does not, so eliminate E.

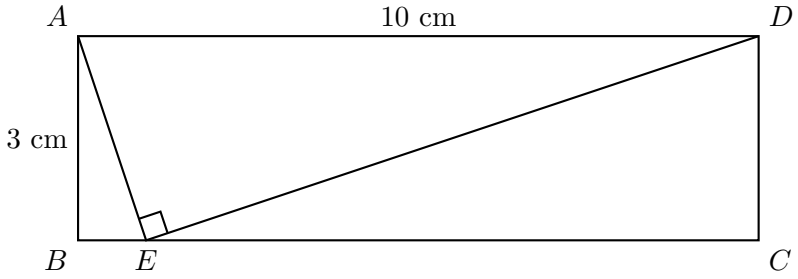
The correct option is therefore **A**, as it is the only one remaining.

Geometry

2. $ABCD$ is a rectangle with $AB = 3$ cm and $AD = 10$ cm.

Point E lies on BC, and $\angle AED = 90°$

What is the value of $BE \times EC$?



A 9 cm^2

B 18 cm^2

C 41 cm^2

D 50 cm^2

E 82 cm^2

Solution **Answer A**

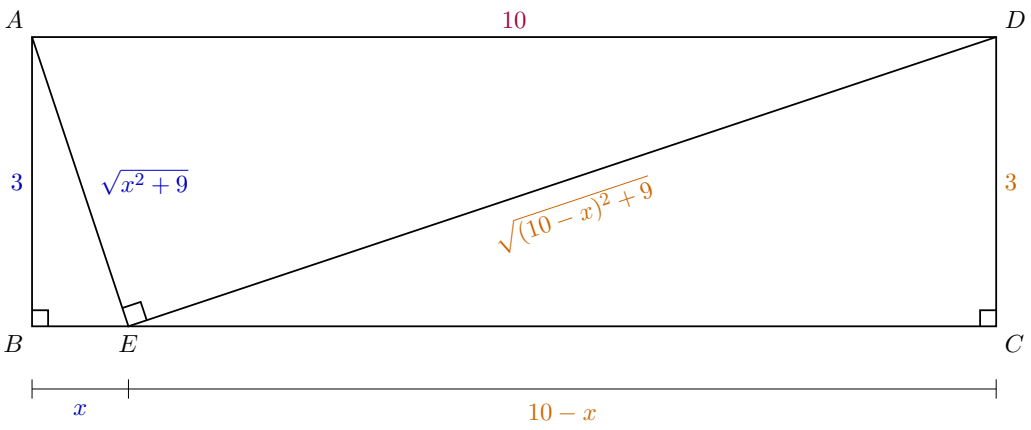
Measure lengths in centimetres and write $BE = x$. Since opposite sides of a rectangle are equal, $BC = AD = 10$ and $CD = AB = 3$, so $EC = 10 - x$.

Applying Pythagoras' theorem in ABE and ECD gives

$$AE^2 = AB^2 + BE^2 = 3^2 + x^2 = x^2 + 9$$

$$DE^2 = CD^2 + EC^2 = 3^2 + (10 - x)^2 = (10 - x)^2 + 9$$

Taking positive square roots gives the lengths labelled below.



Now apply Pythagoras' theorem in AED, where AD is the hypotenuse:

$$AD^2 = AE^2 + DE^2 \implies 10^2 = (x^2 + 9) + \big((10 - x)^2 + 9\big)$$

Expanding and collecting terms,

$$\begin{aligned} 100 &= x^2 + 9 + 100 - 20x + x^2 + 9 \\ &= 2x^2 - 20x + 118 \end{aligned}$$

Rearranging and dividing by 2,

$$x^2 - 10x + 9 = 0$$

Factorising,

$$(x - 1)(x - 9) = 0 \implies x = 1 \text{ or } x = 9$$

Both values place E on BC. The required product is $x(10 - x)$, so either value gives the same result

$$BE \times EC = 1 \times 9 = \boxed{9 \text{ cm}^2}$$

so the correct option is **A**.

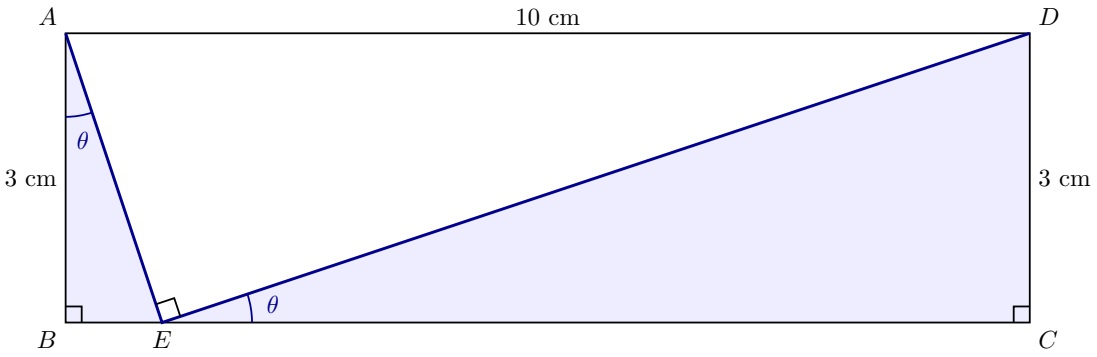
There is also a shorter finish: the quadratic equation gives the product without finding x:

$$x^2 - 10x + 9 = 0 \implies 10x - x^2 = 9 \implies x(10 - x) = 9$$

This is exactly $BE \times EC$

Alternative Solution: (Similar triangles)

We can obtain the product directly by showing that the two smaller right-angled triangles are similar. Write $\theta = \angle BAE$. We will show that $\angle DEC = \theta$ too, as marked below.



Since $\angle ABE = 90°$, the angle sum in triangle ABE gives

$$\angle AEB = 180° - 90° - \theta = 90° - \theta$$

At E, the angles AEB, AED and DEC lie along the straight line BC, so they sum to 180°. Using $\angle AED = 90°$,

$$\angle DEC = 180° - 90° - (90° - \theta) = \theta$$

Thus triangles ABE and ECD are similar: both have a right angle, and $\angle BAE = \angle DEC$. The corresponding vertices are A and E, B and C, then E and D.

Corresponding sides are in the same ratio. Since AB corresponds to EC, and BE to CD,

$$\frac{AB}{EC} = \frac{BE}{CD} \implies \frac{3}{EC} = \frac{BE}{3}$$

Multiplying by $3EC$, which is positive,

$$BE \times EC = 9 \text{ cm}^2$$

Again obtaining option **A**.

Exponentials & Logarithms

3. For $1 < x < 81$, the equation

$$\log_2(\log_3 x) + \log_2\left(\log_3 \frac{81}{x}\right) = 1$$

has two solutions. What is the smaller solution?

A $3^{2-\sqrt{3}}$

B $3^{2-\sqrt{2}}$

C $3^{\sqrt{2}}$

D $3^{4-\sqrt{2}}$

E $3^{2+\sqrt{2}}$

Solution **Answer B**

Both inner logarithms are positive for $1 < x < 81$, so the logarithm rules apply.

Combining the two base-2 logarithms gives

$$\log_2\left[(\log_3 x)\left(\log_3 \frac{81}{x}\right)\right] = 1$$

Since $81 = 3^4$,

$$\begin{aligned}\log_3 \frac{81}{x} &= \log_3 81 - \log_3 x \\ &= 4 - \log_3 x\end{aligned}$$

Substituting gives

$$\log_2\left[(\log_3 x)(4 - \log_3 x)\right] = 1$$

Exponentiating to base 2 gives

$$(\log_3 x)(4 - \log_3 x) = 2$$

This is a hidden quadratic in $\log_3 x$. Put

$$u = \log_3 x$$

Since $\log_3 x$ is increasing, the domain $1 < x < 81$ becomes

$$0 < u < 4$$

The equation is now

$$u(4 - u) = 2$$

Expanding,

$$4u - u^2 = 2$$

Rearranging,

$$u^2 - 4u + 2 = 0$$

The quadratic formula gives

$$u = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(2)}}{2(1)}$$

$$= \frac{4 \pm \sqrt{8}}{2}$$

$$= \frac{4 \pm 2\sqrt{2}}{2}$$

Hence

$$u = 2 - \sqrt{2} \quad \text{or} \quad u = 2 + \sqrt{2}$$

Both values lie in the required domain, because $0 < \sqrt{2} < 2$ gives

$$0 < 2 - \sqrt{2} < 2 + \sqrt{2} < 4$$

Since $\log_3 x$ is increasing, the smaller value of u corresponds to the smaller value of x. Using $x = 3^u$, the smaller solution is

$$x = 3^{2-\sqrt{2}}$$

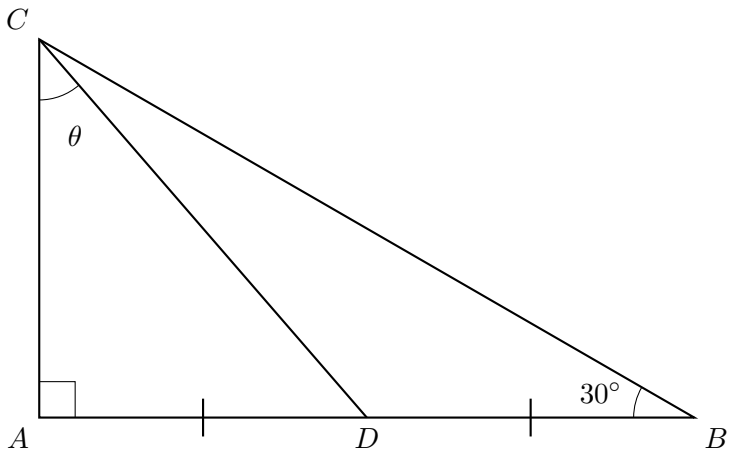
The correct option is **B**.

Trigonometry

4. Triangle ABC is right-angled at A, and angle ABC is $30°$

The point D is the midpoint of AB

If angle ACD is θ, what is $\cos\theta$?



A $\dfrac{1}{\sqrt{7}}$

B $\dfrac{1}{2}$

C $\dfrac{\sqrt{3}}{\sqrt{7}}$

D $\dfrac{2}{\sqrt{7}}$

E $\dfrac{\sqrt{3}}{2}$

Solution **Answer D**

Let $AD = x$, where $x > 0$. Since D is the midpoint of AB, we have $DB = x$ and hence $AB = 2x$

First find AC using triangle ABC:

$$\tan 30° = \frac{AC}{2x}$$

Using $\tan 30° = 1/\sqrt{3}$,

$$\frac{AC}{2x} = \frac{1}{\sqrt{3}}$$

Therefore

$$AC = \frac{2x}{\sqrt{3}}$$

Now find CD in the right-angled triangle ACD. By Pythagoras' theorem,

$$CD^2 = AC^2 + AD^2$$

Substituting,

$$CD^2 = \left(\frac{2x}{\sqrt{3}}\right)^2 + x^2$$

$$= \frac{4x^2}{3} + x^2 = \frac{7x^2}{3}$$

Since $x > 0$ and CD is a length,

$$CD = \frac{\sqrt{7}\,x}{\sqrt{3}}$$

To complete the side labels on the diagram, use triangle ABC:

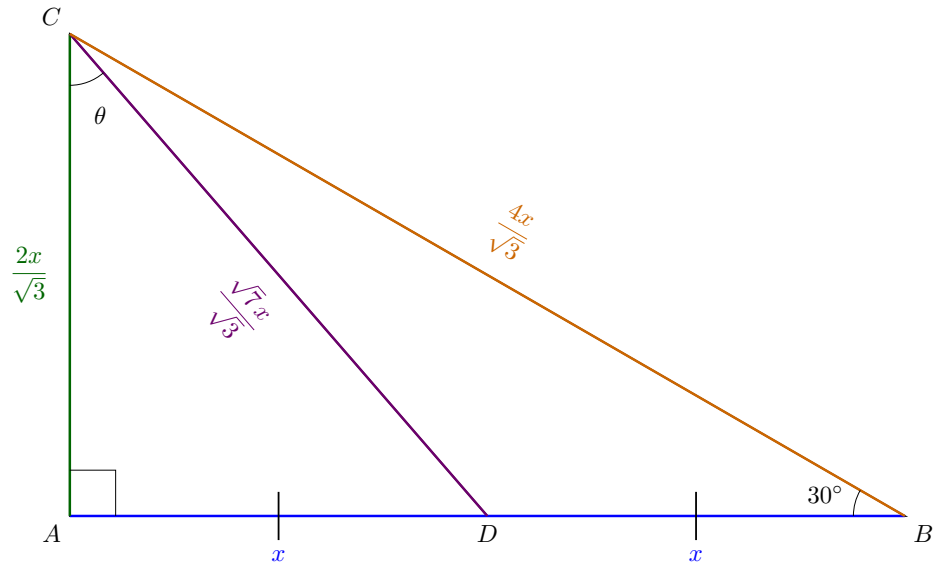
$$\sin 30^\circ = \frac{AC}{BC}$$

Rearranging and substituting,

$$BC = \frac{AC}{\sin 30^\circ}$$

$$= \frac{2x/\sqrt{3}}{1/2} = \frac{4x}{\sqrt{3}}$$

The diagram now shows all the side lengths in terms of x.



For θ in triangle ACD, the adjacent side is AC and the hypotenuse is CD, so

$$\cos\theta = \frac{AC}{CD}$$

Substituting and cancelling the non-zero scale factor x,

$$\cos\theta = \frac{2x/\sqrt{3}}{\sqrt{7}\,x/\sqrt{3}}$$

$$= \frac{2x}{\sqrt{7}\,x} = \frac{2}{\sqrt{7}}$$

The correct option is **D**.

Sequences & Series

5. An infinite geometric series has positive first term a and common ratio r, where $|r| < 1$. The sum to infinity of this series is S.

A second series is formed by taking every third term of the first series, beginning with the first term:

$$T = u_1 + u_4 + u_7 + u_{10} + \dots$$

where u_n denotes the nth term of the first series.

What is the maximum possible value of $\dfrac{T}{S}$?

A $\dfrac{3}{4}$

B 1

C $\dfrac{7}{6}$

D $\dfrac{4}{3}$

E $\dfrac{3}{2}$

F 3

Solution **Answer D**

Write the opening terms of the two series:

$$S = a + ar + ar^2 + ar^3 + \cdots$$

$$T = a + ar^3 + ar^6 + ar^9 + \cdots$$

Both have first term a, but their common ratios are r and r^3, respectively. Since $|r| < 1$, we also have $|r^3| < 1$, so both sums converge.

The geometric sum-to-infinity formula is

$$\text{Sum to infinity} = \frac{\text{first term}}{1 - \text{common ratio}}$$

Applying it to each series gives

$$S = \frac{a}{1 - r}$$

$$T = \frac{a}{1 - r^3}$$

Since $a > 0$ and $r < 1$, we have $S > 0$. Forming the ratio gives

$$\frac{T}{S} = \frac{a}{1 - r^3} \times \frac{1 - r}{a}$$

$$= \frac{1 - r}{1 - r^3} = \frac{r - 1}{r^3 - 1}$$

Since $r^3 - 1 = 0$ at $r = 1$, the factor theorem tells us that $r - 1$ is a factor of the denominator and can be cancelled when $r \neq 1$. Polynomial division of $r^3 - 1$ by $r - 1$ gives quotient

$r^2 + r + 1$ with no remainder, so

$$r^3 - 1 = (r-1)(r^2 + r + 1)$$

Alternatively, this follows from the difference of two cubes identity

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

by taking $a = r$ and $b = 1$.

As $r \neq 1$, cancelling gives

$$\frac{T}{S} = \frac{r-1}{(r-1)(r^2+r+1)}$$

$$= \frac{1}{r^2+r+1}$$

To maximise a reciprocal with a positive denominator, minimise the denominator: dividing 1 by a smaller positive number gives a larger value. Complete the square:

$$r^2 + r + 1 = \left(r + \frac{1}{2}\right)^2 - \frac{1}{4} + 1$$

$$= \left(r + \frac{1}{2}\right)^2 + \frac{3}{4}$$

The squared term is non-negative, so the denominator is always positive and has minimum value

$$\frac{3}{4}$$

Equality occurs when

$$r = -\frac{1}{2}$$

This value is allowed because

$$-1 < -\frac{1}{2} < 1$$

Checking this is essential: the minimum over all real values of r must occur within the permitted range for the corresponding maximum to be attainable.

Hence the maximum possible ratio is

$$\frac{T}{S} = \frac{1}{\frac{3}{4}} = \frac{4}{3}$$

The correct option is **D**.

Exponentials & Logarithms

6. Consider the equation

$$x^{\log_2(x^2)} - 18x^{\log_2 x} + 32 = 0$$

What is the sum of all real solutions to this equation?

A 1

B $\frac{9}{2}$

C 6

D $\frac{25}{4}$

E $\frac{13}{2}$

F $\frac{27}{4}$

G $\frac{31}{4}$

H 18

Solution **Answer F**

The logarithm $\log_2 x$ requires

$$x > 0$$

For a positive argument, the logarithmic power law is $\log_2(t^k) = k \log_2 t$. Applying it here gives

$$\log_2(x^2) = 2 \log_2 x$$

Using the index law $a^{2b} = (a^b)^2$, the first term becomes

$$\begin{aligned} x^{\log_2(x^2)} &= x^{2 \log_2 x} \\ &= \left(x^{\log_2 x}\right)^2 \end{aligned}$$

To turn the equation into a quadratic, set

$$u = x^{\log_2 x} > 0$$

The original equation is then

$$u^2 - 18u + 32 = 0$$

The numbers 2 and 16 have product 32 and sum 18, so factorising gives

$$(u - 2)(u - 16) = 0$$

Hence

$$u = 2 \quad \text{or} \quad u = 16$$

Both values are positive, but these are values of u, not the requested values of x. We must solve $x^{\log_2 x} = u$ for each one.

Taking base-2 logarithms and applying the power law gives

$$\begin{aligned}\log_2 u &= \log_2\left(x^{\log_2 x}\right)\\ &= (\log_2 x)(\log_2 x)\\ &= (\log_2 x)^2\end{aligned}$$

By the definition of logarithms, $\log_2 x = k$ means $x = 2^k$.

For $u = 2$, we obtain

$$(\log_2 x)^2 = \log_2 2 = 1$$

Retaining both square roots gives

$$\log_2 x = 1 \quad \text{or} \quad \log_2 x = -1$$

Converting back to x,

$$x = 2^1 = 2 \quad \text{or} \quad x = 2^{-1} = \frac{1}{2}$$

For $u = 16$, we obtain

$$(\log_2 x)^2 = \log_2 16 = 4$$

Again retaining both square roots gives

$$\log_2 x = 2 \quad \text{or} \quad \log_2 x = -2$$

Converting back to x,

$$x = 2^2 = 4 \quad \text{or} \quad x = 2^{-2} = \frac{1}{4}$$

The complete solution set for x is therefore

$$\boxed{x = 2,\ \frac{1}{2},\ 4,\ \frac{1}{4}}$$

All four values are positive and distinct. Their sum is

$$\begin{aligned}2 + \frac{1}{2} + 4 + \frac{1}{4} &= \frac{8 + 2 + 16 + 1}{4}\\ &= \frac{27}{4}\end{aligned}$$

The correct option is **F**.

Differentiation

7. A straight line is tangent to both the curve $y = x^2$ and the curve $y = \dfrac{1}{x}$, where $x \neq 0$

At what value of x does this line cross the x-axis?

A -2

B $-\dfrac{3}{2}$

C -1

D $-\dfrac{1}{2}$

E $-\dfrac{1}{4}$

Solution **Answer C**

Use separate parameters for the points of contact: a common tangent need not touch the two curves at the same x-coordinate. Let the contacts be (a, a^2) and $(b, 1/b)$, where $b \neq 0$.

For $y = x^2$, the derivative is $2x$, so the tangent at (a, a^2) is

$$y - a^2 = 2a(x - a) \implies y = 2ax - a^2$$

For $y = 1/x$, the derivative is $-1/x^2$, so the tangent at $(b, 1/b)$ is

$$y - \frac{1}{b} = -\frac{1}{b^2}(x - b) \implies y = -\frac{x}{b^2} + \frac{2}{b}$$

For these to be the same line, both the gradients and the y-intercepts must agree:

$$2a = -\frac{1}{b^2} \qquad \text{and} \qquad -a^2 = \frac{2}{b}$$

The first condition gives $a = -1/(2b^2)$. Substituting into the second and using $b \neq 0$,

$$-\frac{1}{4b^4} = \frac{2}{b} \implies -1 = 8b^3 \implies b = -\frac{1}{2}$$

Hence

$$a = -\frac{1}{2\left(-\frac{1}{2}\right)^2} = -2$$

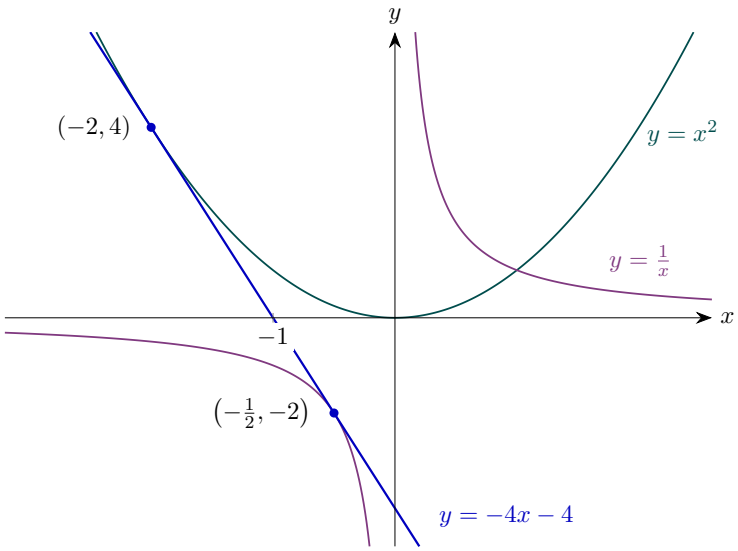
The common tangent is therefore

$$y = 2(-2)x - (-2)^2 = -4x - 4$$

At the x-intercept, $y = 0$, so

$$0 = -4x - 4 \implies \boxed{x = -1}$$

So the correct option is **C**. The sketch shows the two contacts and the x-intercept. The contacts have different x-coordinates.



Alternative Solution:

Let the common tangent be $y = mx + c$. Tangency means that the intersection equation for each curve has a repeated root.

The two intersection equations are

$$x^2 = mx + c \implies x^2 - mx - c = 0$$

$$\frac{1}{x} = mx + c \implies mx^2 + cx - 1 = 0$$

Multiplying the second equation by x introduces no root at $x = 0$, since its constant term is -1. Also, $m \neq 0$: the reciprocal curve has derivative $-1/x^2 \neq 0$, so it has no horizontal tangent.

Both equations are therefore quadratic. Setting their discriminants to zero gives

$$m^2 + 4c = 0 \qquad \text{and} \qquad c^2 + 4m = 0$$

The first condition gives $c = -m^2/4$. Substituting into the second,

$$\frac{m^4}{16} + 4m = 0 \implies m^4 + 64m = 0 \implies m(m^3 + 64) = 0$$

Since $m \neq 0$,

$$m^3 = -64 \implies m = -4 \qquad \text{and} \qquad c = -\frac{(-4)^2}{4} = -4$$

Thus the common tangent is $y = -4x - 4$. Setting $y = 0$ gives

$$0 = -4x - 4 \implies x = -1$$

Again obtaining option **C**.

Sequences & Series

8. Five positive numbers are consecutive terms of a geometric sequence

The middle three have sum 14 and multiply to make 8

What is the sum of all five numbers?

A 74

B 78

C 80

D 82

E 84

F 86

Solution **Answer D**

Putting the middle term at the centre of the notation makes the product of the middle three easy to use. Write the five terms as

$$\frac{b}{r^2}, \quad \frac{b}{r}, \quad b, \quad br, \quad br^2$$

where $b > 0$ and the common ratio $r > 0$. The product of the middle three terms is

$$\frac{b}{r} \times b \times br = 8$$

Cancelling the factors of r gives

$$b^3 = 8 \implies \boxed{b = 2}$$

Now use the sum of the middle three terms:

$$\frac{2}{r} + 2 + 2r = 14 \implies r + \frac{1}{r} = 6$$

There is no need to find r itself: the outer two terms require $r^2 + \frac{1}{r^2}$ which we can obtain by squaring this result:

$$\left(r + \frac{1}{r}\right)^2 = r^2 + 2r\left(\frac{1}{r}\right) + \frac{1}{r^2}$$

So

$$36 = r^2 + 2 + \frac{1}{r^2} \implies \boxed{r^2 + \frac{1}{r^2} = 34}$$

The sum of all five terms is therefore

$$\frac{b}{r^2} + \frac{b}{r} + b + br + br^2 = b\left(\frac{1}{r^2} + \frac{1}{r} + 1 + r + r^2\right)$$

$$= 2(34 + 6 + 1) = \boxed{82}$$

The correct option is **D**.

Alternative Solution:

Avoiding the tricks used in the solution above, we can instead find the common ratio and then calculate each term. Write the five terms as

$$a, \quad ar, \quad ar^2, \quad ar^3, \quad ar^4$$

where $a > 0$ and $r > 0$.

The product of the middle three gives

$$(ar)(ar^2)(ar^3) = a^3r^6 = 8$$

so, taking cube roots,

$$ar^2 = 2 \quad \Longrightarrow \quad a = \frac{2}{r^2}$$

Substituting into their sum gives

$$\frac{2}{r} + 2 + 2r = 14$$

Multiplying by r and rearranging,

$$r^2 - 6r + 1 = 0$$

The quadratic formula gives

$$r = \frac{6 \pm \sqrt{36 - 4}}{2} = 3 \pm 2\sqrt{2}$$

Take $r = 3 + 2\sqrt{2}$. The first term is

$$a = \frac{2}{(3 + 2\sqrt{2})^2} = \frac{2}{17 + 12\sqrt{2}} = \frac{2(17 - 12\sqrt{2})}{289 - 288} = 34 - 24\sqrt{2}$$

Multiplying successively by r gives

$$\begin{aligned} ar &= (34 - 24\sqrt{2})(3 + 2\sqrt{2}) = 6 - 4\sqrt{2} \\ ar^2 &= (6 - 4\sqrt{2})(3 + 2\sqrt{2}) = 2 \\ ar^3 &= 2(3 + 2\sqrt{2}) = 6 + 4\sqrt{2} \\ ar^4 &= (6 + 4\sqrt{2})(3 + 2\sqrt{2}) = 34 + 24\sqrt{2} \end{aligned}$$

Since $(3 + 2\sqrt{2})(3 - 2\sqrt{2}) = 1$, the other ratio gives these same five terms in reverse order, so their sum is unchanged.

Adding, the surd terms cancel:

$$\begin{aligned} a + ar + ar^2 + ar^3 + ar^4 &= (34 - 24\sqrt{2}) + (6 - 4\sqrt{2}) + 2 \\ &\quad + (6 + 4\sqrt{2}) + (34 + 24\sqrt{2}) \\ &= 34 + 6 + 2 + 6 + 34 \\ &= 82 \end{aligned}$$

The correct option is **D**.

Probability & Counting

9. A bag contains 3 red balls and 2 blue balls.

Two balls are drawn without replacement.

If they are the same colour, both are returned to the bag. If they are different colours, both are left out.

A ball is then drawn from the bag.

What is the probability that this final ball is red?

A $\frac{3}{5}$

B $\frac{19}{30}$

C $\frac{16}{25}$

D $\frac{2}{3}$

E $\frac{7}{10}$

Solution **Answer C**

Use a probability tree, keeping track of the balls left in the bag. Multiply along a route to find its probability, and add the probabilities of the routes we want.

First find the probabilities that the initial two balls have the same colour or different colours. There are 5 balls for the first draw and 4 for the second, since the first ball is not replaced.

For the same colour, the two possible routes are red then red and blue then blue:

$$\text{Probability of same colour} = \frac{3}{5} \times \frac{2}{4} + \frac{2}{5} \times \frac{1}{4} = \frac{2}{5}$$

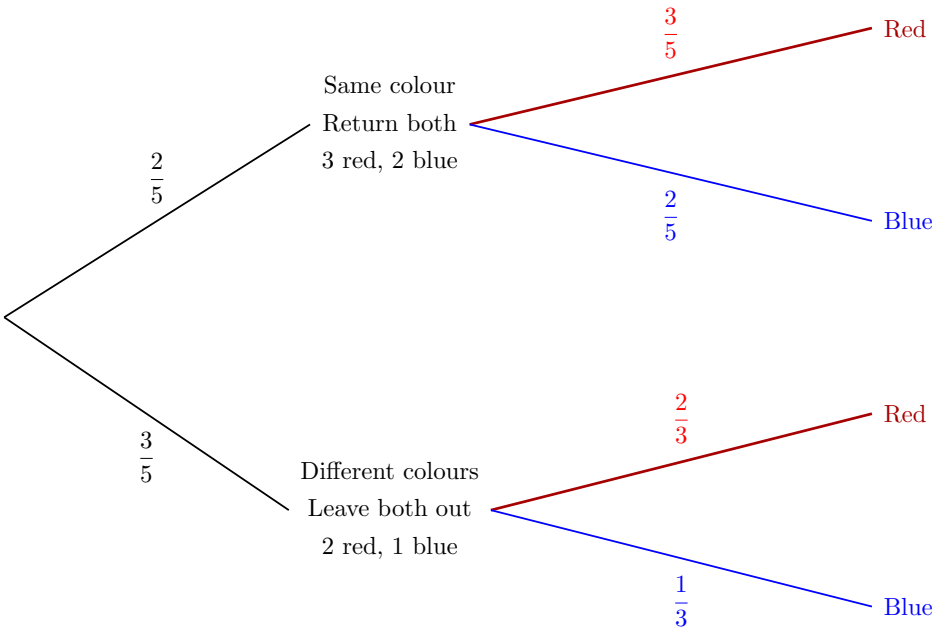
For different colours, the routes are red then blue and blue then red:

$$\text{Probability of different colours} = \frac{3}{5} \times \frac{2}{4} + \frac{2}{5} \times \frac{3}{4} = \frac{3}{5}$$

These give the first branches of the tree below. To complete the tree, apply the rule in the question:

- Same colour: return both balls, so the bag again contains 3 red and 2 blue.
- Different colours: leave both out, so the bag contains 2 red and 1 blue.

The final branches use these new bag contents.



There are two routes ending in red. Multiply along each complete route:

$$\text{Same colour, then final red:} \quad \frac{2}{5} \times \frac{3}{5} = \frac{6}{25}$$

$$\text{Different colours, then final red:} \quad \frac{3}{5} \times \frac{2}{3} = \frac{2}{5}$$

Add these to obtain the probability that the final ball is red:

$$\frac{6}{25} + \frac{2}{5} = \frac{6}{25} + \frac{10}{25} = \boxed{\frac{16}{25}}$$

The correct option is **C**.

Algebra

10. Real numbers x and y satisfy

$$x + y = xy$$

and

$$(x-1)^2 + (y-1)^2 = 7$$

What are the possible values of xy?

A -5 and 1

B -3 and 3

C -1 and 3

D -1 and 5

E 1 and 5

Solution **Answer D**

The required product is also the sum, so let t denote their common value:

$$t = x + y = xy$$

First expand the second equation:

$$x^2 + y^2 - 2(x+y) + 2 = 7$$

To express this in terms of t, use the identity

$$x^2 + y^2 = (x+y)^2 - 2xy$$

which follows by expanding $(x+y)^2 = x^2 + 2xy + y^2$.

Substituting the sum and product gives

$$x^2 + y^2 = t^2 - 2t$$

Hence the expanded second equation becomes

$$t^2 - 2t - 2t + 2 = 7 \implies t^2 - 4t - 5 = 0$$

The numbers -5 and 1 have product -5 and sum -4, so the factorisation is

$$(t-5)(t+1) = 0$$

Therefore the candidate values are

$$t = 5 \quad \text{or} \quad t = -1$$

Thus

$$\boxed{xy = 5 \quad \text{or} \quad xy = -1}$$

Option **D** is now the only possible answer, although a complete proof would also check that both values arise from real numbers x and y satisfying the original equations.

We can, in principle, instead explicitly solve for x and y and multiply to find xy. However this direct route involves painful algebra and, at the very least, is unsuitable for the pace demanded by the TMUA.

Probability & Counting

11. Find the coefficient of x^3y^2 in the expansion of

$$(1-2x+y)^8$$

A -17920

B -13440

C -4480

D -560

E 560

F 4480

G 17920

Solution **Answer C**

Regard $(1-2x+y)^8$ as a product of eight identical brackets. Each bracket supplies one term, so to obtain x^3y^2 we must choose $-2x$ from three brackets, y from two brackets, and 1 from the remaining three.

Recall that the number of ways to choose r objects from n possibilities, without regard to order, is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Choose the three brackets supplying $-2x$, then the two supplying y from the five left. The remaining brackets must supply 1, so the number of selections is

$$\binom{8}{3}\binom{5}{2} = \frac{8\times 7\times 6}{3\times 2\times 1}\times\frac{5\times 4}{2\times 1} = 56\times 10 = 560$$

Each selection contributes $(-2x)^3y^2 = -8x^3y^2$. Therefore the required coefficient is

$$560\times(-8) = \boxed{-4480}$$

The correct option is **C**.

Alternative Solution 1:

Apply the binomial theorem twice, first treating $1-2x$ as one term. Recall that

$$(a+b)^n = \sum_{r=0}^{n}\binom{n}{r}a^{n-r}b^r$$

With $a = 1-2x$, $b = y$ and $n = 8$,

$$\big((1-2x)+y\big)^8 = \sum_{r=0}^{8}\binom{8}{r}(1-2x)^{8-r}y^r$$

Only $r = 2$ can give y^2, since the remaining power contains no y. The relevant term is therefore

$$\binom{8}{2}(1-2x)^6 y^2$$

Now apply the binomial theorem to the remaining power:

$$(1-2x)^6 = \sum_{r=0}^{6} \binom{6}{r} 1^{6-r}(-2x)^r$$

Only $r = 3$ gives x^3, with coefficient $\binom{6}{3}(-2)^3$. Hence the coefficient of x^3y^2 in the original expansion is

$$\binom{8}{2}\binom{6}{3}(-2)^3 = 28 \times 20 \times (-8) = -4480$$

Again obtaining option **C**.

Alternative Solution 2:

The trinomial theorem gives a short calculation here, but is not needed for either of the preceding methods. Note that the trinomial theorem is far beyond the TMUA specification.

For a non-negative integer n, it states that

$$(a+b+c)^n = \sum_{i+j+k=n} \frac{n!}{i!j!k!} a^i b^j c^k$$

The sum runs over all non-negative integers i, j and k whose total is n. These are the respective exponents of a, b and c. A zero exponent is allowed, with $0! = 1$.

Substituting $a = 1$, $b = -2x$, $c = y$ and $n = 8$ gives

$$(1-2x+y)^8 = \sum_{i+j+k=8} \frac{8!}{i!j!k!} 1^i (-2x)^j y^k$$

To obtain x^3y^2, we need $j = 3$ and $k = 2$, so $i = 8-3-2 = 3$. The required coefficient is therefore

$$\frac{8!}{3!3!2!}(-2)^3 = \frac{8 \times 7 \times 6 \times 5 \times 4}{(3 \times 2 \times 1)(2 \times 1)}(-8) = 560(-8) = -4480$$

The correct option is **C**.

Remark: (Optional)

The trinomial theorem extends the binomial theorem from two terms to three. More generally, the *multinomial theorem* applies to a sum of m terms:

$$(a_1 + \cdots + a_m)^n = \sum_{i_1+\cdots+i_m=n} \frac{n!}{i_1! \cdots i_m!} a_1^{i_1} \cdots a_m^{i_m}$$

Here m is the number of terms in the bracket, and n is a non-negative integer. The sum runs over all non-negative integer exponents $i_1, \ldots, i_m$ whose total is n.

Each exponent records how many of the n brackets supply the corresponding term. The factorial expression counts the ways to make those selections.

Taking $m = 3$, with $a_1 = a$, $a_2 = b$ and $a_3 = c$, gives the trinomial theorem. Taking $m = 2$ gives

$$(a+b)^n = \sum_{i+j=n} \frac{n!}{i!j!} a^i b^j$$

Since $i + j = n$, choosing j fixes $i = n - j$. We can therefore write the sum using j alone:

$$(a+b)^n = \sum_{j=0}^{n} \frac{n!}{(n-j)!j!} a^{n-j} b^j = \sum_{j=0}^{n} \binom{n}{j} a^{n-j} b^j$$

The summation index is a dummy variable: its name does not change the sum. Renaming j as r gives exactly the binomial theorem quoted above:

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$$

Thus the multinomial theorem includes both the binomial and trinomial theorems.

Inequalities

12. The function f is defined for all real x by

$$f(x) = \frac{x^2 + 4x + 1}{x^2 + 1}$$

Which of the following gives all possible values of $f(x)$ as x varies over the real numbers?

A $-1 \leq f(x) \leq 3$

B $f(x) \leq -1$ or $f(x) \geq 3$

C $1 \leq f(x) \leq 3$

D $-3 \leq f(x) \leq 1$

E $-1 \leq f(x) \leq 1$

F $f(x)$ can take all real values

Solution **Answer A**

Put $y = f(x)$ and find which values of y allow a real input x. Since $x^2 + 1 > 0$ for every real x, multiplying by the denominator gives

$$y(x^2 + 1) = x^2 + 4x + 1$$

Rearranging as an equation in x,

$$(y - 1)x^2 - 4x + (y - 1) = 0$$

First suppose $y \neq 1$, so this is a quadratic. A quadratic $ax^2 + bx + c = 0$, with $a \neq 0$, has at least one real root exactly when its discriminant satisfies

$$b^2 - 4ac \geq 0$$

Here the discriminant is

$$\begin{aligned} b^2 - 4ac &= (-4)^2 - 4(y - 1)(y - 1) \\ &= 16 - 4(y - 1)^2 \end{aligned}$$

Therefore a real input exists exactly when

$$16 - 4(y - 1)^2 \geq 0$$

Rearranging,

$$(y - 1)^2 \leq 4$$

Taking the bounds on $y - 1$,

$$-2 \leq y - 1 \leq 2$$

Adding 1,

$$-1 \leq y \leq 3$$

The endpoints are included because a zero discriminant gives a repeated real root. At either endpoint, that root is

$$x = \frac{-(-4)}{2(y-1)}$$

Substituting the endpoint values,

$$y = -1: \quad x = \frac{4}{2(-2)} = -1$$

$$y = 3: \quad x = \frac{4}{2(2)} = 1$$

Now check $y = 1$ separately, since the quadratic coefficient then vanishes. Substituting into the rearranged equation gives

$$-4x = 0$$

Hence

$$x = 0$$

So $y = 1$ is also attained. Every value between -1 and 3 gives a real solution of the rearranged equation, and multiplying by the positive denominator was reversible, so every such solution is an input with $f(x) = y$.

No value outside these bounds gives a real input. Thus all possible values are

$$-1 \leq f(x) \leq 3$$

The correct option is **A**.

Alternative Solution:

This approach uses rational-function analysis and the quotient rule, which are beyond the TMUA specification but might be familiar to some candidates.

The question asks for the *range* of f: all output values it can take, or equivalently all heights reached by its graph.

There are no vertical asymptotes because $x^2 + 1 > 0$. The numerator and denominator have the same degree and leading coefficient, so the horizontal asymptote at both ends is $y = 1$.

Also,

$$f(x) = 1 + \frac{4x}{x^2+1}$$

shows that the graph lies below $y = 1$ for $x < 0$, above it for $x > 0$, and crosses it at $(0, 1)$.

For a quotient $y = u(x)/v(x)$, the quotient rule states that

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{u'(x)v(x) - u(x)v'(x)}{[v(x)]^2}$$

Applying it here,

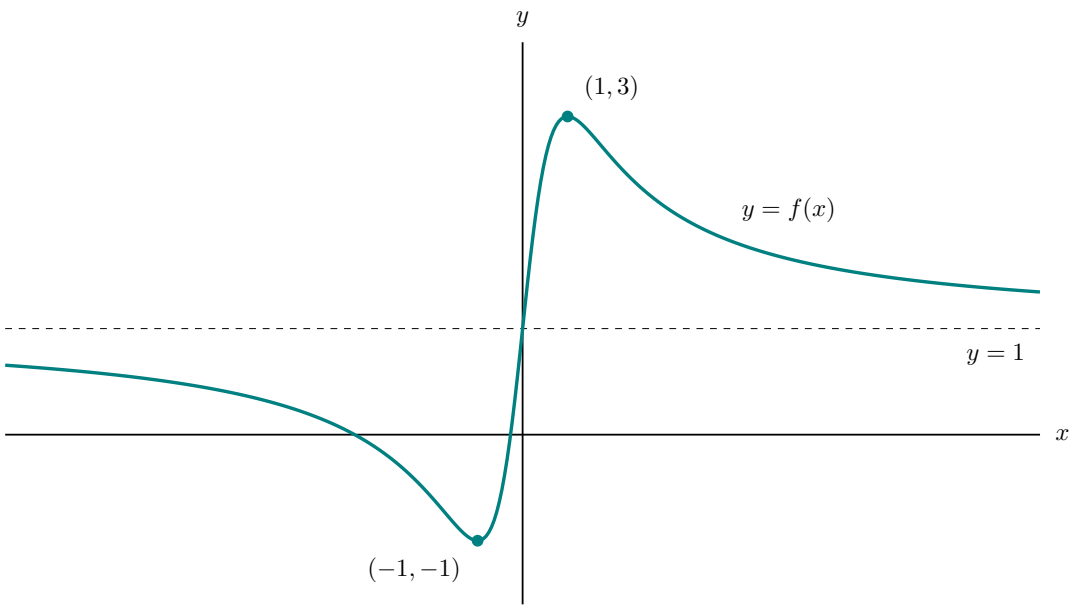
$$f'(x) = \frac{(x^2+1)(2x+4) - (x^2+4x+1)(2x)}{(x^2+1)^2}$$

$$= \frac{4(1-x^2)}{(x^2+1)^2}$$

Hence

$$f'(x) = 0 \implies x = \pm 1 \qquad \text{with} \qquad f(-1) = -1, \quad f(1) = 3$$

The derivative is negative for $x < -1$, positive for $-1 < x < 1$, and negative for $x > 1$. Together with the horizontal asymptote, this gives the following sketch.



The minimum is -1, the maximum is 3, and continuity ensures that every value between them occurs. Therefore the range is

$$-1 \leq f(x) \leq 3$$

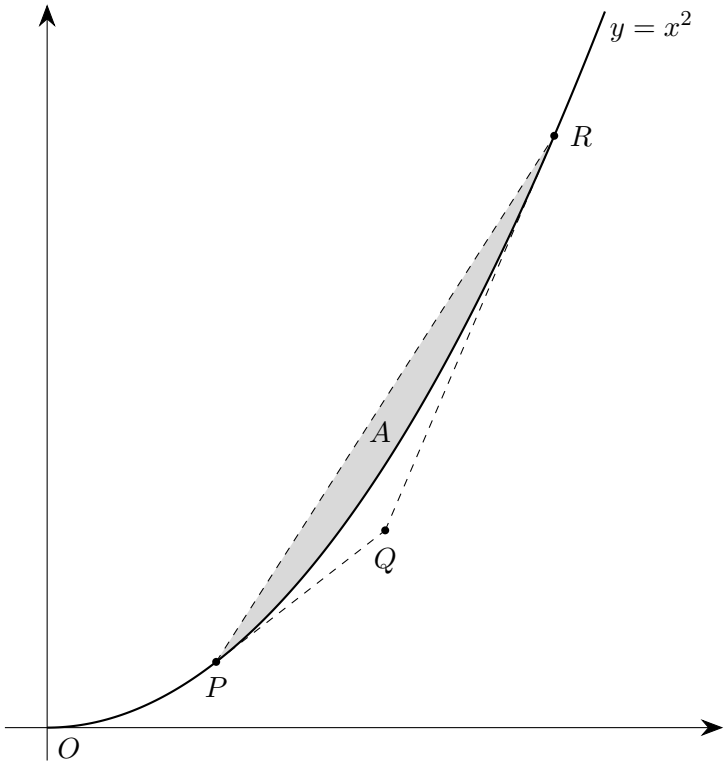
The correct option is **A**.

Coordinate Geometry

13. The points P and R lie on the curve $y = x^2$ and have x-coordinates 1 and 3, respectively. The tangents to the curve at P and R meet at Q.

The area enclosed by the curve and the line segment PR is A. The area of triangle PQR is B.

What is the value of $\dfrac{A}{B}$?



A $\dfrac{1}{3}$

B $\dfrac{1}{2}$

C $\dfrac{2}{3}$

D $\dfrac{3}{4}$

E $\dfrac{3}{2}$

Solution **Answer C**

Substituting into $y = x^2$ gives

$$P = (1, 1), \qquad R = (3, 9)$$

The gradient of PR is $(9 - 1)/(3 - 1) = 4$, so its equation is

$$y - 1 = 4(x - 1) \implies y = 4x - 3$$

Since $\mathrm{d}y/\mathrm{d}x = 2x$, the tangent gradients at P and R are 2 and 6. Hence

$$\text{At } P: \quad y - 1 = 2(x-1) \implies y = 2x - 1$$

$$\text{At } R: \quad y - 9 = 6(x-3) \implies y = 6x - 9$$

At their intersection,

$$2x - 1 = 6x - 9 \implies x = 2, \qquad y = 3$$

so $Q = (2, 3)$.

First calculate A. The chord lies above the curve between P and R, so integrate upper minus lower:

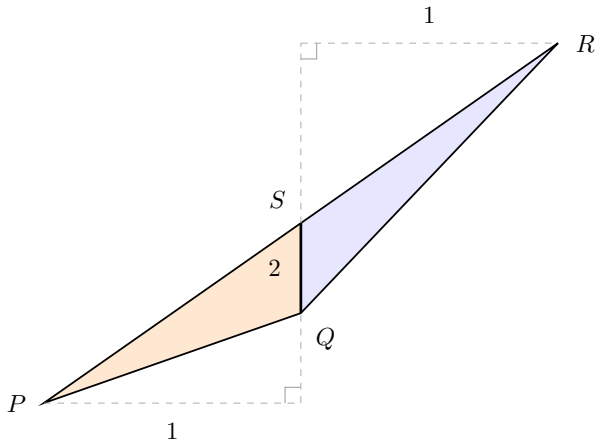
$$A = \int_1^3 \left((4x-3) - x^2\right) \mathrm{d}x$$

$$= \left[2x^2 - 3x - \frac{x^3}{3}\right]_1^3$$

$$= 0 - \left(-\frac{4}{3}\right) = \frac{4}{3}$$

For B, draw the vertical segment from Q to the chord, meeting it at S. Substituting $x = 2$ into $y = 4x - 3$ gives $S = (2, 5)$, so $QS = 2$.

The diagram below isolates triangle PQR. The two shaded triangles share the vertical base QS. Their perpendicular heights are the horizontal distances from P and R to the line $x = 2$, both equal to 1.



$B =$ area of $PQS +$ area of RQS

Using area $= \frac{1}{2} \times$ base $\times$ perpendicular height for each triangle,

$$B = \frac{1}{2} \times 2 \times 1 + \frac{1}{2} \times 2 \times 1 = 2$$

Therefore

$$\frac{A}{B} = \frac{\frac{4}{3}}{2} = \frac{2}{3}$$

The correct option is **C**.

Alternative Solution: (Other formulae for B)

The following triangle-area formulae are beyond the specification. We reuse $P = (1,1)$, $Q = (2,3)$, $R = (3,9)$ and $A = \frac{4}{3}$ from the primary solution.

For two side vectors $\begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ starting at the same vertex, the triangle's area is

$$\text{area} = \frac{1}{2}\left|u_1 v_2 - u_2 v_1\right|$$

This is half the magnitude of their cross product. Here $\overrightarrow{PQ} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\overrightarrow{PR} = \begin{pmatrix} 2 \\ 8 \end{pmatrix}$, so

$$B = \frac{1}{2}\left|1 \times 8 - 2 \times 2\right| = 2$$

Alternatively, for vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3), the *shoelace formula* is

$$\text{area} = \frac{1}{2}\left|(x_1 y_2 + x_2 y_3 + x_3 y_1) - (y_1 x_2 + y_2 x_3 + y_3 x_1)\right|$$

Using the vertices in the order P, Q, R,

$$B = \frac{1}{2}\left|(1 \times 3 + 2 \times 9 + 3 \times 1) - (1 \times 2 + 3 \times 3 + 9 \times 1)\right| = \frac{1}{2}\left|24 - 20\right| = 2$$

Either calculation gives

$$\frac{A}{B} = \frac{\frac{4}{3}}{2} = \frac{2}{3}$$

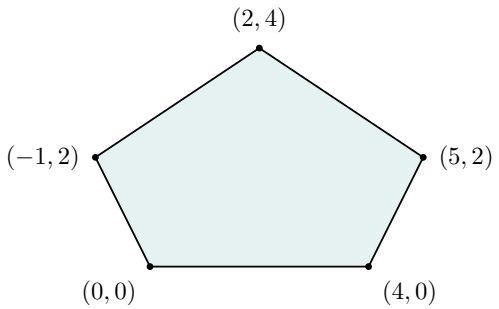
The correct option is **C**.

Remark: (Optional, the shoelace formula)

The shoelace formula works for any polygon whose sides do not cross. List its vertices $(x_1, y_1), \ldots, (x_n, y_n)$ in order around the boundary, clockwise or anticlockwise, and repeat the first vertex at the end. Then

$$\text{area} = \frac{1}{2}\left|\sum_{i=1}^{n} x_i y_{i+1} - \sum_{i=1}^{n} y_i x_{i+1}\right| \qquad (x_{n+1}, y_{n+1}) = (x_1, y_1)$$

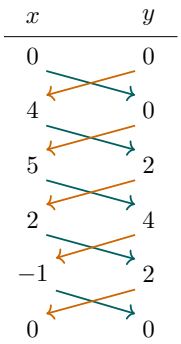
For example, consider this pentagon.



Write the coordinates in boundary order, repeating the first pair:

x	y
0	0
4	0
5	2
2	4
-1	2
0	0

Multiply each left-hand entry by the next right-hand entry and add; then do the same in the opposite direction.



These crossing products give the formula its name:

$$\begin{aligned}\text{area} &= \frac{1}{2}\Big|\big(0(0)+4(2)+5(4)+2(2)+(-1)(0)\big)\\ &\qquad -\big(0(4)+0(5)+2(2)+4(-1)+2(0)\big)\Big|\\ &= \frac{1}{2}|32-0| = 16\end{aligned}$$

So the area of the pentagon is 16.

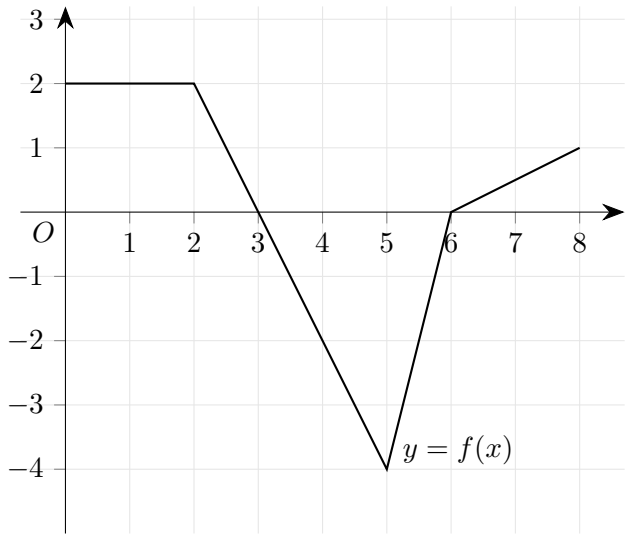
Integration

14. The diagram shows the graph of $y = f(x)$ for $0 \leq x \leq 8$. The graph is made up of straight line segments joining the points $(0, 2)$, $(2, 2)$, $(5, -4)$, $(6, 0)$ and $(8, 1)$.

The function g is defined by

$$g(x) = \int_0^x f(t)\,\mathrm{d}t \qquad \text{for } 0 \leq x \leq 8$$

How many values of x with $0 \leq x \leq 8$ satisfy $g(x) = 0$?



A 0

B 1

C 2

D 3

E 4

F 5

Solution **Answer D**

$g(x)$ is the accumulated signed area from 0 to x: area above the axis is added, and area below it is subtracted. We want the points where these areas balance.

First, $g(0) = 0$, so $x = 0$ is one solution.

The area of a trapezium is

$$\text{area} = \frac{1}{2}(a + b)h$$

where a and b are the lengths of the parallel sides and h is the perpendicular distance between them.

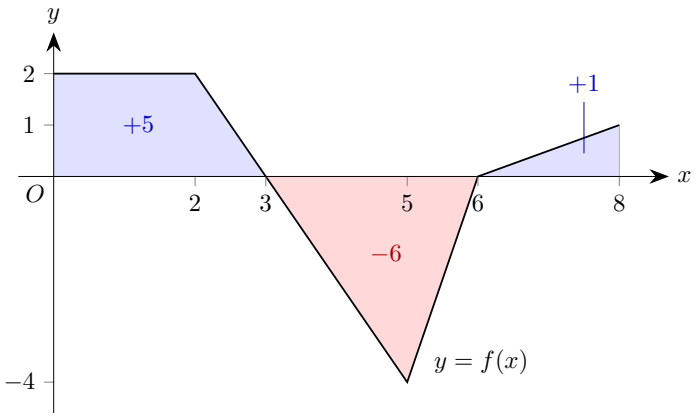
From 0 to 3, the positive region is a trapezium with parallel sides of lengths 2 and 3,

separated by a distance of 2. Hence

$$g(3) = \frac{1}{2}(2+3) \times 2 = 5$$

Only positive area is added over this interval, so there is no further zero before $x = 3$.

The diagram summarises the three signed areas.

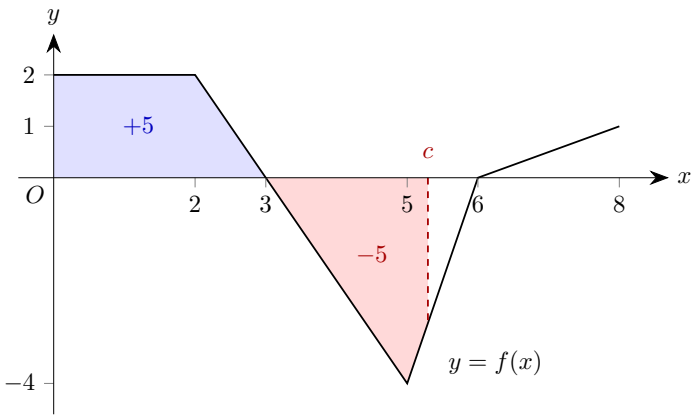


From 3 to 5, the area subtracted is $\frac{1}{2} \times 2 \times 4 = 4$, leaving

$$g(5) = 5 - 4 = 1$$

From 5 to 6, the total area available to subtract is $\frac{1}{2} \times 1 \times 4 = 2$. Only an area of 1 is needed to cancel the remaining positive balance.

As we move right from 5 to 6, the area subtracted grows from 0 to 2, so it reaches 1 exactly once. Call that position c, as shown in the sketch:



After c, more negative area is added, giving

$$g(6) = 1 - 2 = -1$$

Finally, the positive triangle from 6 to 8 has area $\frac{1}{2} \times 2 \times 1 = 1$. All of this area is needed to cancel the deficit, so the balance returns to zero only at $x = 8$.

There are therefore 3 solutions: $x = 0$, one value between 5 and 6, and $x = 8$. The correct option is **D**.

Number

15. Let k be a positive integer, and let $N = 2^4 \times 3^k$

The product of all the positive factors of N is the fourth power of an integer.

What is the smallest possible value of k?

A 1

B 2

C 3

D 5

E 7

F 8

Solution **Answer E**

Every positive factor has the form $2^i 3^j$, where

$$0 \leq i \leq 4, \qquad 0 \leq j \leq k$$

Thus the product of all the factors can be written in five rows:

$$\begin{aligned}
&\left(2^0 3^0 \times 2^0 3^1 \times \cdots \times 2^0 3^k\right) \\
&\times \left(2^1 3^0 \times 2^1 3^1 \times \cdots \times 2^1 3^k\right) \\
&\times \left(2^2 3^0 \times 2^2 3^1 \times \cdots \times 2^2 3^k\right) \\
&\times \left(2^3 3^0 \times 2^3 3^1 \times \cdots \times 2^3 3^k\right) \\
&\times \left(2^4 3^0 \times 2^4 3^1 \times \cdots \times 2^4 3^k\right)
\end{aligned}$$

Each exponent of 2 occurs $k+1$ times, while each exponent of 3 occurs five times.

Using the arithmetic series formula $0 + 1 + \cdots + k = \frac{1}{2}k(k+1)$,

$$\text{Total exponent of } 2 = (k+1)(0+1+2+3+4) = 10(k+1)$$

$$\text{Total exponent of } 3 = 5(0 + 1 + \cdots + k) = \frac{5}{2}k(k+1)$$

Hence the product is

$$2^{10(k+1)} 3^{\frac{5}{2}k(k+1)}$$

An integer is a fourth power exactly when every prime exponent is divisible by 4.

For $10(k+1)$ to be divisible by 4, $k+1$ must be even, so k is odd. For $\frac{5}{2}k(k+1)$ to be divisible by 4, we need $k(k+1)$ to be divisible by 8.

Since k is odd, all three required factors of 2 must come from $k+1$. Thus $k+1$ must be a multiple of 8, giving

$$k = 7,\ 15,\ 23,\ \ldots$$

At $k = 7$, the product is indeed a fourth power:

$$2^{80}3^{140} = \left(2^{20}3^{35}\right)^4$$

The smallest possible value is $k = 7$, so the correct option is **E**.

Alternatively, once the prime exponents are known, test the options in increasing order. For $k = 1, 2, 3, 5$, an exponent is respectively $5, 30, 30, 75$, none divisible by 4. For $k = 7$, the exponents are 80 and 140, both divisible by 4, so the first listed value that works gives option **E**.

Alternative Solution:

There are five choices for the exponent of 2 and $k + 1$ choices for the exponent of 3, giving $5(k + 1)$ positive factors.

Match each factor d with N/d. Each pair has product N.

If k is odd, N is not a square, so all factors form distinct pairs:

$$\text{Product of all factors} = N^{\frac{5}{2}(k+1)}$$

If k is even, the factor $\sqrt{N}$ is left over after pairing the others. Including it once gives the same result:

$$\text{Product of all factors} = N^{\frac{5}{2}(k+1)-\frac{1}{2}}\sqrt{N} = N^{\frac{5}{2}(k+1)}$$

In either case,

$$\text{Product of all factors} = (2^4 \times 3^k)^{\frac{5}{2}(k+1)} = 2^{10(k+1)}3^{\frac{5}{2}k(k+1)}$$

This is the same expression as in the primary solution, so the same divisibility argument gives $k = 7$ and option **E**.

Polynomials

16. A quadratic polynomial f has roots 1 and 3.

The equation $f(f(x)) = 0$ has exactly three distinct real solutions.

What is the value of $f(0)$?

A -9

B -3

C -1

D 3

E 9

Solution **Answer A**

The roots determine the factors, so write

$$f(x) = a(x-1)(x-3), \qquad a \neq 0$$

Including a is essential: knowing the roots tells us where the graph crosses the x-axis, but not its vertical scale or whether it opens upwards or downwards. Writing only $(x-1)(x-3)$ would assume $a = 1$ without justification. The extra condition determines a.

Since the outer f vanishes only at 1 and 3,

$$f(f(x)) = 0 \iff f(x) = 1 \quad \text{or} \quad f(x) = 3$$

These equations have no solutions in common. Each has at most two real solutions, so exactly three distinct solutions require one horizontal line to touch the parabola at its vertex and the other to cut it twice.

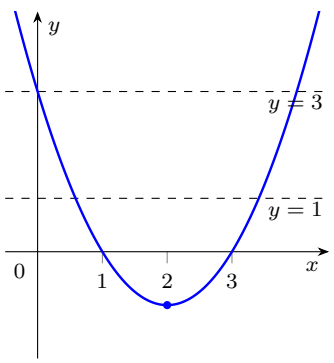
Completing the square,

$$f(x) = a(x^2 - 4x + 3) = a(x-2)^2 - a$$

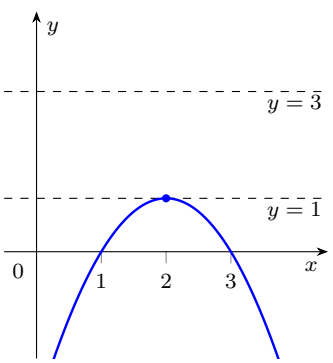
so the vertex is $(2, -a)$.

The sketches show $y = f(x)$ in blue, together with the levels $y = 1$ and $y = 3$.

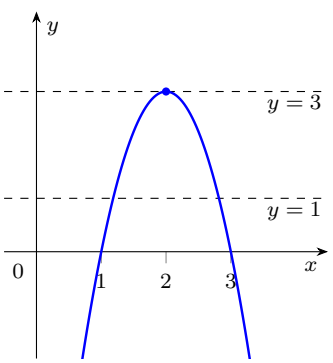
$a > 0$



$a = -1$



$a = -3$



If $a > 0$, the vertex lies below the x-axis, so both horizontal lines cut the parabola twice. This gives four solutions.

If $a < 0$, the vertex is a maximum. A vertex at height 1 gives only one solution: $y = 1$ touches once and $y = 3$ misses the curve. A vertex at height 3 gives exactly three: $y = 3$ touches once and $y = 1$ cuts twice.

Therefore

$$-a = 3 \implies a = -3$$

and hence

$$f(0) = -3(0-1)(0-3) = -9$$

The correct option is **A**.

Alternative Solution:

Using $f(x) = a(x-1)(x-3)$, the equations $f(x) = 1$ and $f(x) = 3$ become

$$ax^2 - 4ax + 3a - 1 = 0$$

$$ax^2 - 4ax + 3a - 3 = 0$$

Their solution sets are disjoint. To give three distinct real solutions, one quadratic must have discriminant zero and the other must have positive discriminant.

Using $b^2 - 4ac$, their respective discriminants are

$$\Delta_1 = (-4a)^2 - 4a(3a-1) = 4a(a+1)$$

$$\Delta_3 = (-4a)^2 - 4a(3a-3) = 4a(a+3)$$

Since $a \neq 0$, the two possibilities are

$$\Delta_1 = 0 \implies a = -1 \implies \Delta_3 = -8 < 0$$

$$\Delta_3 = 0 \implies a = -3 \implies \Delta_1 = 24 > 0$$

Only $a = -3$ gives one repeated root and two further distinct real roots. Thus

$$f(0) = -3(0-1)(0-3) = -9$$

The correct option is **A**.

Trigonometry

17. A triangle has an angle of $120°$

The perimeter of the triangle is 15, and its area is $\frac{15\sqrt{3}}{4}$

What is the length of the longest side of the triangle?

A 5

B $\frac{11}{2}$

C 6

D $\frac{13}{2}$

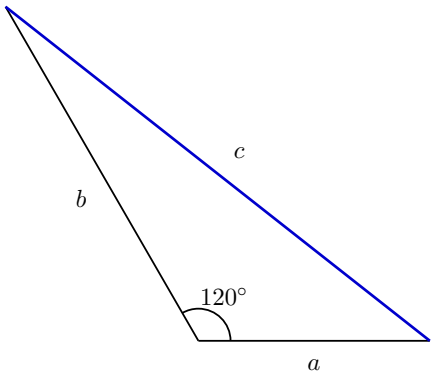
E 7

F $\frac{15}{2}$

G 8

Solution **Answer E**

Let the sides adjacent to the $120°$ angle be a and b, and let the opposite side be c.



The other two angles sum to $60°$, so $120°$ is the largest angle. The longest side is opposite the largest angle, so we need to find c.

Using the area formula area $= \frac{1}{2}ab\sin\theta$ and $\sin 120° = \frac{\sqrt{3}}{2}$,

$$\frac{1}{2}ab\sin 120° = \frac{ab\sqrt{3}}{4} = \frac{15\sqrt{3}}{4} \implies ab = 15$$

The perimeter gives

$$a + b + c = 15 \implies a + b = 15 - c$$

By the cosine rule, with $\cos 120° = -\frac{1}{2}$,

$$\begin{aligned} c^2 &= a^2 + b^2 - 2ab\cos 120° \\ &= a^2 + b^2 + ab \end{aligned}$$

We know $a+b$ and ab, rather than a and b separately, so use $a^2+b^2=(a+b)^2-2ab$. Substituting $a+b=15-c$ and $ab=15$ gives

$$c^2=(15-c)^2-15$$

Expanding and cancelling c^2 from both sides,

$$c^2=225-30c+c^2-15$$

$$0=210-30c$$

Hence

$$30c=210 \implies c=7$$

This also satisfies the triangle inequality for the longest side, since $c=7<8=a+b$.

Thus the longest side has length 7, and the correct option is **E**.

Coordinate Geometry

18. The point A has coordinates $(-7, -5)$ and the point B has coordinates $(23, 45)$.

How many points on the straight line segment AB have both coordinates equal to integers, not counting A and B themselves?

A 4

B 5

C 9

D 10

E 11

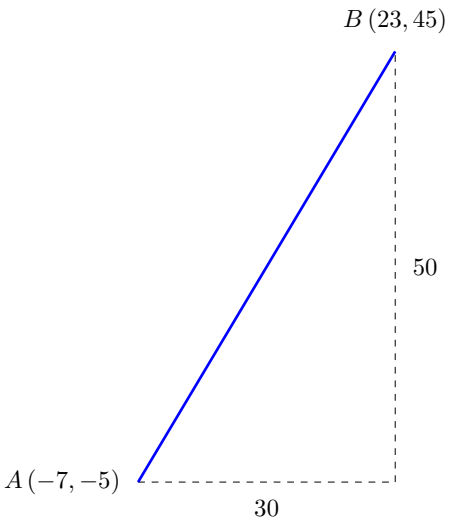
F 29

G 49

Solution **Answer C**

The horizontal and vertical changes from A to B are

$$23 - (-7) = 30 \qquad \text{and} \qquad 45 - (-5) = 50$$



Since $30 : 50 = 3 : 5$, each step $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ takes us from one integer-coordinate point to another on the line.

To see that these steps miss no such points, the line equation $y - y_1 = m(x - x_1)$ gives

$$y + 5 = \frac{5}{3}(x + 7) \implies 3(y + 5) = 5(x + 7)$$

If x and y are integers, $5(x + 7)$ must be divisible by 3. Since 3 and 5 have no common factor, $x + 7 = 3k$ for some integer k. The line equation then gives

$$y + 5 = \frac{5}{3}(3k) = 5k$$

Hence every integer-coordinate point on the line has the form

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -7 \\ -5 \end{pmatrix} + k \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

so it lies a whole number of these steps from A.

Now

$$\overrightarrow{AB} = \begin{pmatrix} 30 \\ 50 \end{pmatrix} = 10 \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

so A corresponds to $k = 0$ and B to $k = 10$. The interior points therefore correspond to $k = 1, 2, \ldots, 9$, giving $\boxed{9}$ points.

The correct option is **C**.

Algebra

19. The positive real numbers x and y satisfy the equations

$$x\sqrt{y} + y\sqrt{x} = 8$$

and

$$x\sqrt{x} + y\sqrt{y} = 40$$

What is the value of $x + y$?

A 6

B 8

C 10

D 12

E 14

F 16

Solution **Answer D**

The square roots suggest the substitution

$$a = \sqrt{x}, \qquad b = \sqrt{y}$$

where a and b are positive. Then $x = a^2$ and $y = b^2$, so the given equations become

$$a^2b + ab^2 = 8$$

$$a^3 + b^3 = 40$$

Factorising the first equation gives

$$ab(a + b) = 8$$

To find $a + b$, expand its cube directly:

$$\begin{aligned}(a+b)^3 &= (a^2 + 2ab + b^2)(a + b) \\ &= a^3 + 3a^2b + 3ab^2 + b^3 \\ &= (a^3 + b^3) + 3(a^2b + ab^2)\end{aligned}$$

This expansion combines the second given equation with three times the first. Substituting their values gives

$$(a + b)^3 = 40 + 3(8) = 64$$

Taking the cube root,

$$\boxed{a + b = 4}$$

Substituting into the factored first equation,

$$4ab = 8$$

Dividing by 4,

$$\boxed{ab = 2}$$

We do not need to find a and b separately, because expanding the square of their sum gives

$$(a+b)^2 = a^2 + 2ab + b^2$$

Rearranging and substituting,

$$\begin{aligned} a^2 + b^2 &= (a+b)^2 - 2ab \\ &= 4^2 - 2(2) \\ &= 12 \end{aligned}$$

Therefore

$$x + y = 12$$

The correct option is **D**.

Remark:

We could instead use $a + b = 4$ and $ab = 2$ to find a and b separately:

$$a(4-a) = 2 \implies a^2 - 4a + 2 = 0 \implies a = 2 \pm \sqrt{2}$$

Then $b = 4 - a$ is the other value, so

$$x + y = (2+\sqrt{2})^2 + (2-\sqrt{2})^2 = 12$$

This works, but finding a and b individually is unnecessary.

Integration

20. Let f be a polynomial of degree at most 3 with real coefficients.

For $t > 0$, define

$$D(t) = \int_0^t f(x)\,\mathrm{d}x - \int_{-t}^0 f(x)\,\mathrm{d}x$$

Which of the following statements are true?

I If $f(x) = f(-x)$ for every real x, then $D(1) = 0$

II If $D(1) = 0$, then $f(x) = f(-x)$ for every real x

III If $D(1) = D(2) = 0$, then $f(x) = f(-x)$ for every real x

A None of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Solution **Answer F**

The condition $f(x) = f(-x)$ for every real x is the definition of an *even function*. Even functions are precisely those for which the y-axis is a line of symmetry.

A polynomial is an even function precisely when it contains only even powers of x. This includes the constant term, which we view as the x^0 term. To see this, write

$$f(x) = ax^3 + bx^2 + cx + k$$

where the coefficients may be zero, so this includes every polynomial of degree at most 3.

Replacing x by $-x$ gives

$$f(-x) = -ax^3 + bx^2 - cx + k$$

For this to equal $f(x)$ for every real x, corresponding coefficients must agree:

$$a = -a, \qquad c = -c \implies a = c = 0$$

Thus the condition is exactly that both odd-power coefficients vanish, as we expected for the definition of an even function applied to a cubic.

Now calculate the two integrals:

$$\int_0^t f(x)\,\mathrm{d}x = \left[\frac{ax^4}{4} + \frac{bx^3}{3} + \frac{cx^2}{2} + kx\right]_0^t$$

$$= \frac{at^4}{4} + \frac{bt^3}{3} + \frac{ct^2}{2} + kt$$

and

$$\int_{-t}^0 f(x)\,\mathrm{d}x = \left[\frac{ax^4}{4} + \frac{bx^3}{3} + \frac{cx^2}{2} + kx\right]_{-t}^0$$

$$= 0 - \left(\frac{at^4}{4} - \frac{bt^3}{3} + \frac{ct^2}{2} - kt\right)$$

$$= -\frac{at^4}{4} + \frac{bt^3}{3} - \frac{ct^2}{2} + kt$$

Subtracting, the contributions from bx^2 and k cancel, giving

$$\boxed{D(t) = \frac{1}{2}at^4 + ct^2}$$

Statement I. If $f(x) = f(-x)$ for every real x, then $a = c = 0$ as discussed above. Hence

$$D(1) = \frac{0}{2} + 0 = 0$$

Statement I is true.

Statement II. The condition $D(1) = 0$ gives

$$\frac{a}{2} + c = 0 \implies c = -\frac{a}{2}$$

This relates the coefficients but does not force both to be zero. For example, take $f(x) = 2x^3 - x$. Then

$$D(1) = \frac{2}{2} - 1 = 0$$

but $f(1) = 1$ and $f(-1) = -1$, so f is not even. Statement II is false.

Statement III. The two conditions give

$$D(1) = 0 \implies \frac{a}{2} + c = 0$$

$$D(2) = 0 \implies 8a + 4c = 0$$

From the first equation, $c = -\frac{a}{2}$. Substituting into the second,

$$8a + 4\left(-\frac{a}{2}\right) = 0 \implies 6a = 0 \implies a = 0$$

Then $c = -\frac{a}{2} = 0$ as well. Both odd-power coefficients vanish, so $f(x) = f(-x)$ for every real x. Statement III is true.

Thus I and III only are true, so the correct option is **F**.